

laborcode.r

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```
setwd("~/Dropbox/Classes/536/CaseStudies/Labor")
rm(list=ls())

#read the data; the file is saved in STATA format, so you need this library:
library(foreign)
data <- read.dta('MROZ.dta');

#the wage of the non-working women is zero, hence the variables
#wage and lwage (columns 7 and 21) will contain missing values
#we eliminate these variables from the data

data = data[,setdiff(colnames(data),c("wage","lwage"))]
#data <- data[,c(1,3:6,8:19,22)]

#transform some variables
data[, "age"] = log(data[, "age"])
data[, "hushrs"] = log(data[, "hushrs"])

#display the names of the variables
colnames(data)

## [1] "inlf"      "hours"     "kidslt6"   "kidsge6"   "age"       "educ"
## [7] "repwage"   "hushrs"    "husage"    "huseduc"    "huswage"    "faminc"
## [13] "mtr"       "motheduc"  "fatheduc"  "unem"       "city"       "exper"
## [19] "nwifeinc"  "expersq"

for(i in 2:18)
{
  a = data[,i]
  data[,i] = (a-mean(a))/sd(a)
}

#now the data contains 753 samples and 19 variables
attach(data)

#fit the model involving all the 18 indendent variables
fullmodel = glm(inlf~.,family=binomial(link=logit),data=data)

## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

summary(fullmodel)

##
## Call:
## glm(formula = inlf ~ ., family = binomial(link = logit), data = data)
##
## Deviance Residuals:
```

```
##           Min           1Q           Median           3Q           Max
## -7.512e-04 -2.000e-08  2.000e-08  2.000e-08  1.122e-03
##
```

```
## Coefficients:
```

```
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)  2.084e+03  4.982e+05  0.004  0.997
## hours        9.455e+02  9.564e+04  0.010  0.992
## kidslt6     -4.400e+00  2.798e+03 -0.002  0.999
## kidsge6      7.354e+00  1.416e+03  0.005  0.996
## age          1.531e+01  3.108e+03  0.005  0.996
## educ         1.413e+01  2.137e+03  0.007  0.995
## repwage     -2.923e+00  3.071e+03 -0.001  0.999
## hushrs      -1.217e+00  2.630e+03  0.000  1.000
## husage      -1.476e+01  4.779e+03 -0.003  0.998
## huseduc     -1.322e+01  1.641e+03 -0.008  0.994
## huswage      1.886e+00  1.414e+03  0.001  0.999
## faminc       7.052e+02  3.041e+05  0.002  0.998
## mtr          1.008e+00  3.845e+03  0.000  1.000
## motheduc    -3.337e+00  1.377e+03 -0.002  0.998
## fatheduc     1.447e+00  8.853e+02  0.002  0.999
## unem        -1.645e-01  7.559e+02  0.000  1.000
## city        -2.606e+00  1.588e+03 -0.002  0.999
## exper       -1.265e+01  2.968e+03 -0.004  0.997
## nwifeinc    -5.778e+01  2.481e+04 -0.002  0.998
## expersq      3.902e-02  8.069e+00  0.005  0.996
```

```
##
## (Dispersion parameter for binomial family taken to be 1)
##
```

```
## Null deviance: 1.0297e+03 on 752 degrees of freedom
## Residual deviance: 4.5196e-06 on 733 degrees of freedom
## AIC: 40
```

```
##
## Number of Fisher Scoring iterations: 25
```

```
#find the correlation of each explanatory variable with the response
for(i in 2:19)
{
  cat(colnames(data)[i], "[", i, "] = ", cor(data[,1], data[,i]), "\n");
}
```

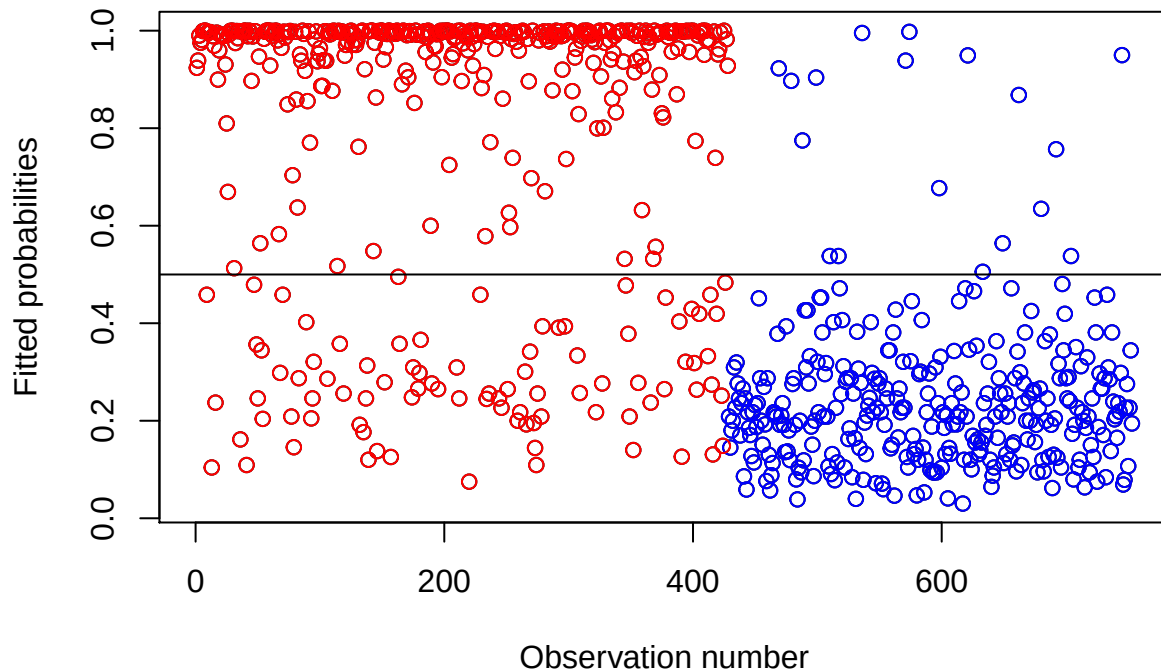
```
## hours [ 2 ] = 0.7411454
## kidslt6 [ 3 ] = -0.2137493
## kidsge6 [ 4 ] = -0.002424231
## age [ 5 ] = -0.07226078
## educ [ 6 ] = 0.1873528
## repwage [ 7 ] = 0.6340485
## hushrs [ 8 ] = -0.06160755
## husage [ 9 ] = -0.07282005
## huseduc [ 10 ] = 0.04591422
## huswage [ 11 ] = -0.06947526
## faminc [ 12 ] = 0.09889538
## mtr [ 13 ] = -0.1448255
## motheduc [ 14 ] = 0.09048973
## fatheduc [ 15 ] = 0.05771841
## unem [ 16 ] = -0.02873489
```

```
## city [ 17 ] = -0.006167593
## exper [ 18 ] = 0.3424847
## nwifeinc [ 19 ] = -0.1175972

#fit a logistic regression with these variables
M0 = glm(inlf~kidslt6+educ+repwage+exper,family=binomial(link=logit),data=data)
summary(M0)

##
## Call:
## glm(formula = inlf ~ kidslt6 + educ + repwage + exper, family = binomial(link = logit),
##      data = data)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -3.4744  -0.6681   0.0195   0.2756   2.2757
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   1.7392     0.2358   7.375 1.65e-13 ***
## kidslt6       -0.3734     0.1220  -3.061 0.002206 **
## educ           0.4702     0.1201   3.914 9.07e-05 ***
## repwage        3.5650     0.3265  10.918 < 2e-16 ***
## exper          0.4279     0.1176   3.638 0.000275 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1029.75  on 752  degrees of freedom
## Residual deviance:  512.18  on 748  degrees of freedom
## AIC: 522.18
##
## Number of Fisher Scoring iterations: 7

#make a plot of the fitted values
myind = 1:length(inlf)
plot(myind,M0$fitted.values,xlab="Observation number",ylab="Fitted probabilities")
points(myind[inlf==0],M0$fitted.values[inlf==0],col="blue")
points(myind[inlf==1],M0$fitted.values[inlf==1],col="red")
abline(h=0.5)
```



```
Mempty = glm(inlf~1,family=binomial(link=logit),data=data)
summary(Mempty)
```

```
##
## Call:
## glm(formula = inlf ~ 1, family = binomial(link = logit), data = data)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -1.296  -1.296   1.063   1.063   1.063
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.27530    0.07358   3.742 0.000183 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1029.7  on 752  degrees of freedom
## Residual deviance: 1029.7  on 752  degrees of freedom
## AIC: 1031.7
##
## Number of Fisher Scoring iterations: 4
```

```
Mfull = glm(inlf~.,family=binomial(link=logit),data=data)
```

```
## Warning: glm.fit: algorithm did not converge
```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
summary(Mfull)
```

```
##
```

```

## Call:
## glm(formula = inlf ~ ., family = binomial(link = logit), data = data)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -7.512e-04 -2.000e-08  2.000e-08  2.000e-08  1.122e-03
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  2.084e+03  4.982e+05   0.004   0.997
## hours        9.455e+02  9.564e+04   0.010   0.992
## kidslt6     -4.400e+00  2.798e+03  -0.002   0.999
## kidsge6      7.354e+00  1.416e+03   0.005   0.996
## age         1.531e+01  3.108e+03   0.005   0.996
## educ        1.413e+01  2.137e+03   0.007   0.995
## repwage     -2.923e+00  3.071e+03  -0.001   0.999
## hushrs      -1.217e+00  2.630e+03   0.000   1.000
## husage     -1.476e+01  4.779e+03  -0.003   0.998
## huseduc    -1.322e+01  1.641e+03  -0.008   0.994
## huswage     1.886e+00  1.414e+03   0.001   0.999
## faminc      7.052e+02  3.041e+05   0.002   0.998
## mtr         1.008e+00  3.845e+03   0.000   1.000
## motheduc   -3.337e+00  1.377e+03  -0.002   0.998
## fatheduc    1.447e+00  8.853e+02   0.002   0.999
## unem       -1.645e-01  7.559e+02   0.000   1.000
## city       -2.606e+00  1.588e+03  -0.002   0.999
## exper      -1.265e+01  2.968e+03  -0.004   0.997
## nwifeinc   -5.778e+01  2.481e+04  -0.002   0.998
## expersq     3.902e-02  8.069e+00   0.005   0.996
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1.0297e+03  on 752  degrees of freedom
## Residual deviance: 4.5196e-06  on 733  degrees of freedom
## AIC: 40
##
## Number of Fisher Scoring iterations: 25
M1 = step(Mempty,trace=TRUE,direction="both",scope=list(upper = Mfull,lower = Mempty))

## Start:  AIC=1031.75
## inlf ~ 1

## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

##           Df Deviance    AIC
## + hours    1     0.00    4.00
## + repwage   1   554.82  558.82
## + exper     1   931.20  935.20
## + expersq   1   966.67  970.67
## + kidslt6   1   994.75  998.75
## + educ      1  1002.69 1006.69
## + mtr       1  1013.56 1017.56
## + nwifeinc  1  1019.31 1023.31

```

```

## + faminc      1  1022.14 1026.14
## + motheduc    1  1023.56 1027.56
## + husage      1  1025.74 1029.74
## + age         1  1025.80 1029.80
## + huswage     1  1026.12 1030.12
## + hushrs      1  1026.84 1030.84
## + fatheduc    1  1027.23 1031.23
## <none>        1029.75 1031.75
## + huseduc     1  1028.16 1032.16
## + unem        1  1029.12 1033.12
## + city        1  1029.72 1033.72
## + kidsge6     1  1029.74 1033.74

## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
## Step: AIC=4
## inlf ~ hours

## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge

```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
##           Df Deviance    AIC
## <none>          0.0    4.0
## + nwifeinc    1      0.0    6.0
## + faminc      1      0.0    6.0
## + exper       1      0.0    6.0
## + mtr         1      0.0    6.0
## + expersq     1      0.0    6.0
## + repwage     1      0.0    6.0
## + unem        1      0.0    6.0
## + educ        1      0.0    6.0
## + huswage     1      0.0    6.0
## + hushrs      1      0.0    6.0
## + kidsge6     1      0.0    6.0
## + motheduc    1      0.0    6.0
## + kidslt6     1      0.0    6.0
## + husage      1      0.0    6.0
## + city        1      0.0    6.0
## + huseduc     1      0.0    6.0
## + age         1      0.0    6.0
## + fatheduc    1      0.0    6.0
## - hours      1 1029.8 1031.8
```

```
summary(M1)
```

```
##
## Call:
## glm(formula = inlf ~ hours, family = binomial(link = logit),
##      data = data)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -1.478e-04 -1.478e-04  2.000e-08  2.000e-08  2.379e-03
```

```
##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)      1902      29604   0.064   0.949
## hours           2259      35082   0.064   0.949
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1.0297e+03 on 752 degrees of freedom
## Residual deviance: 1.2760e-05 on 751 degrees of freedom
## AIC: 4
##
## Number of Fisher Scoring iterations: 25

#now try this model
mylogit <- glm(inlf~hushrs+kidslt6+educ+exper,family=binomial(link=logit))
summary(mylogit)

##
## Call:
## glm(formula = inlf ~ hushrs + kidslt6 + educ + exper, family = binomial(link = logit))
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -2.6717  -1.0311   0.5367   0.9683   2.0347
##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.35257    0.08294   4.251 2.13e-05 ***
## hushrs       -0.11504    0.08610  -1.336   0.181
## kidslt6      -0.41641    0.08700  -4.786 1.70e-06 ***
## educ         0.47131    0.08797   5.358 8.43e-08 ***
## exper        0.76167    0.09879   7.710 1.26e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1029.75 on 752 degrees of freedom
## Residual deviance: 881.61 on 748 degrees of freedom
## AIC: 891.61
##
## Number of Fisher Scoring iterations: 4

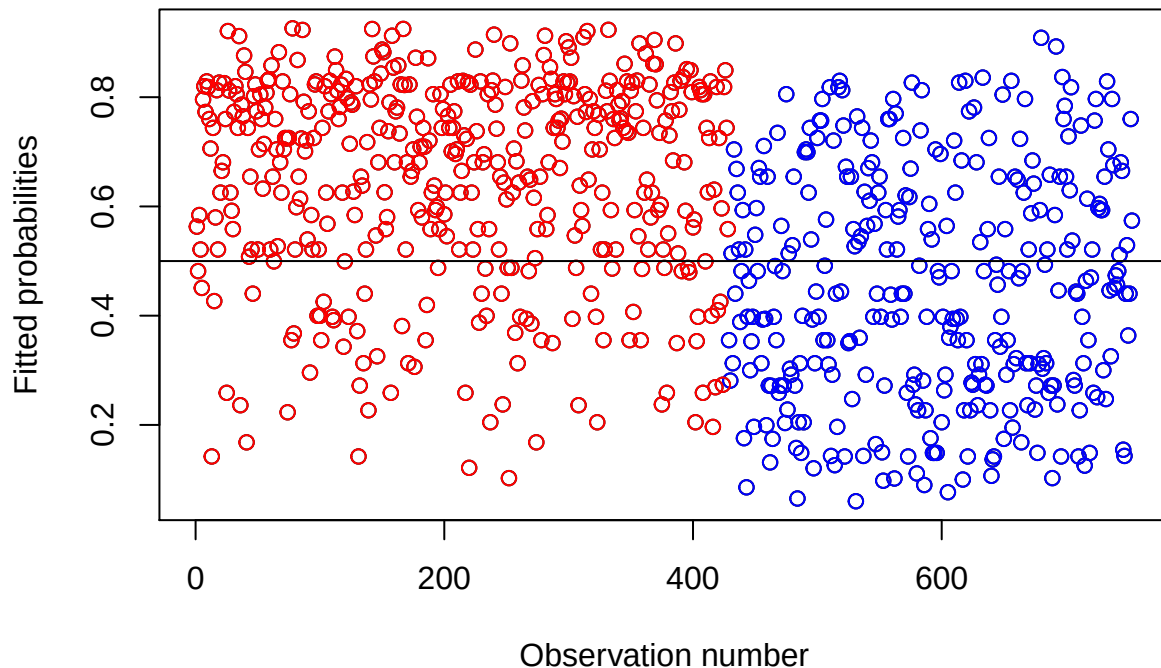
#one more
mylogit <- glm(inlf~kidslt6+educ+exper+expersq,family=binomial(link=logit))
summary(mylogit)

##
## Call:
## glm(formula = inlf ~ kidslt6 + educ + exper + expersq, family = binomial(link = logit))
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -2.1884  -1.0011   0.6097   0.9032   2.1340
##
```



```
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.0243485  0.1895227   5.405 6.48e-08 ***
## kidslt6     -0.4257112  0.0879164  -4.842 1.28e-06 ***
## educ        0.4331045  0.0882786   4.906 9.29e-07 ***
## exper       1.6300679  0.2436415   6.690 2.23e-11 ***
## expersq     -0.0039682  0.0009779  -4.058 4.95e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1029.75  on 752  degrees of freedom
## Residual deviance:  868.88  on 748  degrees of freedom
## AIC: 878.88
##
## Number of Fisher Scoring iterations: 4
```

```
#make a plot of the fitted values
myind = 1:length(inlf)
plot(myind,mylogit$fitted.values,xlab="Observation number",ylab="Fitted probabilities")
points(myind[inlf==0],mylogit$fitted.values[inlf==0],col="blue")
points(myind[inlf==1],mylogit$fitted.values[inlf==1],col="red")
abline(h=0.5)
```

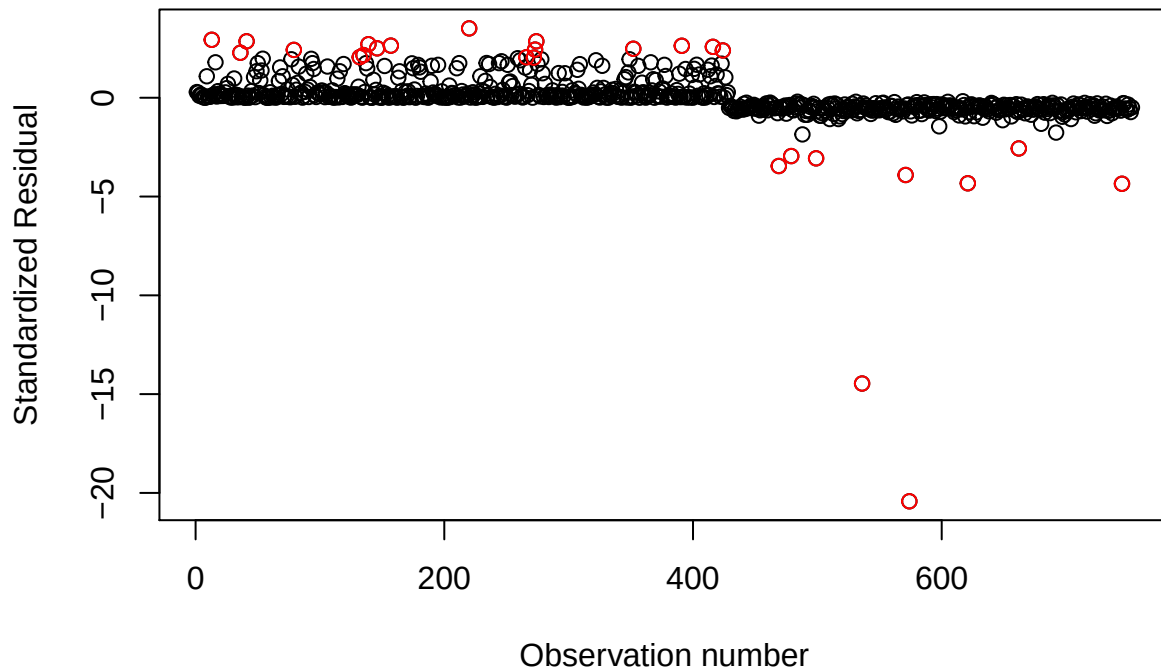


```
#determine the standardized residuals
myres = (inlf-M0$fitted.values)/sqrt(M0$fitted.values*(1-M0$fitted.values))
#calculate the p-value for the chisq test
1-pchisq(sum(myres^2),length(inlf)-length(coef(M0)))
```

```
## [1] 0
```

```
#make an index plot of standardized residuals against observation number
plot(1:length(inlf),myres,xlab="Observation number",ylab="Standardized Residual")
```

```
#the outliers are outside the (-2,2) interval
a = (1:length(inlf))[abs(myres)>=2]
points(a,myres[a],col='red')
```



```
#calculate the Brier Score
(brier = mean((inlf-mylogit$fitted.values)^2))
```

```
## [1] 0.1968666
```

```
#calculate the error rate = proportion of incorrectly predicted samples
(error.rate <- mean((mylogit$fitted.values>0.5 & inlf==0) | (mylogit$fitted.values<0.5 & inlf==1)))
```

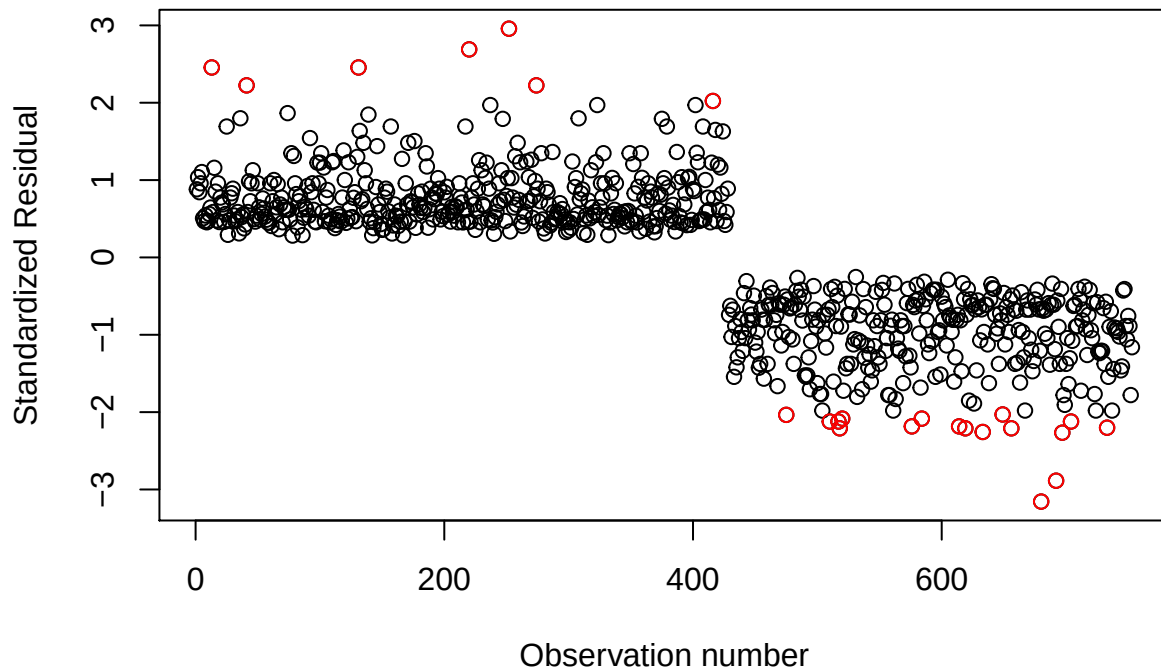
```
## [1] 0.2974768
```

```
#determine the standardized residuals
myres = (inlf-mylogit$fitted.values)/sqrt(mylogit$fitted.values*(1-mylogit$fitted.values))
#calculate the p-value for the chisq test
1-pchisq(sum(myres^2),length(inlf)-length(coef(mylogit)))
```

```
## [1] 0.5227109
```

```
#make an index plot of standardized residuals against observation number
plot(1:length(inlf),myres,xlab="Observation number",ylab="Standardized Residual")
```

```
#the outliers are outside the (-2,2) interval
a = (1:length(inlf))[abs(myres)>=2]
points(a,myres[a],col='red')
```



```
#calculate the aic with the formula
mylogit$deviance+2*mylogit$rank
```

```
## [1] 878.8846
```

```
#you can also get it directly
mylogit$aic
```

```
## [1] 878.8846
```

```
AIC(mylogit)
```

```
## [1] 878.8846
```

```
#calculate the bic
(bic = mylogit$deviance+mylogit$rank*log(length(inlf)))
```

```
## [1] 902.005
```

```
BIC(mylogit)
```

```
## [1] 902.005
```

```
#now use the stepwise function to see if we can improve the model
```

```
library(stats)
mylogit1 <- glm(inlf~.,family=binomial(link=logit),data=data)
```

```
## Warning: glm.fit: algorithm did not converge
```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
summary(mylogit1)
```

```
##
```

```
## Call:
```

```
## glm(formula = inlf ~ ., family = binomial(link = logit), data = data)
```

```
##
```

```

## Deviance Residuals:
##      Min        1Q      Median        3Q        Max
## -7.512e-04 -2.000e-08  2.000e-08  2.000e-08  1.122e-03
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  2.084e+03  4.982e+05  0.004  0.997
## hours        9.455e+02  9.564e+04  0.010  0.992
## kidslt6     -4.400e+00  2.798e+03 -0.002  0.999
## kidsge6      7.354e+00  1.416e+03  0.005  0.996
## age         1.531e+01  3.108e+03  0.005  0.996
## educ        1.413e+01  2.137e+03  0.007  0.995
## repwage     -2.923e+00  3.071e+03 -0.001  0.999
## hushrs      -1.217e+00  2.630e+03  0.000  1.000
## husage     -1.476e+01  4.779e+03 -0.003  0.998
## huseduc    -1.322e+01  1.641e+03 -0.008  0.994
## huswage     1.886e+00  1.414e+03  0.001  0.999
## faminc      7.052e+02  3.041e+05  0.002  0.998
## mtr         1.008e+00  3.845e+03  0.000  1.000
## motheduc   -3.337e+00  1.377e+03 -0.002  0.998
## fatheduc    1.447e+00  8.853e+02  0.002  0.999
## unem       -1.645e-01  7.559e+02  0.000  1.000
## city       -2.606e+00  1.588e+03 -0.002  0.999
## exper      -1.265e+01  2.968e+03 -0.004  0.997
## nwifeinc   -5.778e+01  2.481e+04 -0.002  0.998
## expersq     3.902e-02  8.069e+00  0.005  0.996
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1.0297e+03  on 752  degrees of freedom
## Residual deviance: 4.5196e-06  on 733  degrees of freedom
## AIC: 40
##
## Number of Fisher Scoring iterations: 25
newmodel <- step(mylogit1,trace=TRUE)

## Start:  AIC=40
## inlf ~ hours + kidslt6 + kidsge6 + age + educ + repwage + hushrs +
##      husage + huseduc + huswage + faminc + mtr + motheduc + fatheduc +
##      unem + city + exper + nwifeinc + expersq
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge

```


[illegible]

```

## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
##           Df    Deviance AIC
## - repwage   1 4.3955e-06 36
## - mtr       1 4.4245e-06 36
## - unem      1 4.4331e-06 36
## - huswage   1 4.4661e-06 36
## - fatheduc  1 4.4964e-06 36
## - city      1 4.5600e-06 36
## - motheduc  1 4.6057e-06 36
## - expersq   1 4.9326e-06 36
## - exper     1 5.0306e-06 36
## - kidslt6   1 5.1060e-06 36
## - kidsge6   1 5.3103e-06 36
## - husage    1 5.3939e-06 36
## - age       1 5.9906e-06 36
## - nwifeinc  1 6.0831e-06 36
## - faminc    1 6.0846e-06 36
## - educ      1 6.5597e-06 36
## - huseduc   1 6.6257e-06 36
## - hours     1 1.0210e-05 36
## <none>      4.4399e-06 38
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
## Step: AIC=36
## inlf ~ hours + kidslt6 + kidsge6 + age + educ + husage + huseduc +
##       huswage + faminc + mtr + motheduc + fatheduc + unem + city +
##       exper + nwifeinc + expersq
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

```



```

## - motheduc 1 4.5763e-06 34
## - expersq 1 4.7683e-06 34
## - exper 1 4.7704e-06 34
## - kidsge6 1 4.9747e-06 34
## - kidslt6 1 5.0123e-06 34
## - husage 1 5.3155e-06 34
## - age 1 5.8381e-06 34
## - faminc 1 6.0868e-06 34
## - nwifeinc 1 6.1028e-06 34
## - huseduc 1 6.2309e-06 34
## - educ 1 6.3851e-06 34
## - hours 1 1.0127e-05 34
## <none> 4.3955e-06 36

## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
## Step: AIC=34
## inlf ~ hours + kidslt6 + kidsge6 + age + educ + husage + huseduc +
## huswage + faminc + motheduc + fatheduc + unem + city + exper +
## nwifeinc + expersq

## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge

```

```

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
##      Df    Deviance AIC
## - unem      1 4.3610e-06 32
## - huswage    1 4.3676e-06 32
## - fatheduc    1 4.4473e-06 32
## - city        1 4.4913e-06 32
## - motheduc    1 4.5376e-06 32
## - expersq      1 4.7268e-06 32
## - exper        1 4.7538e-06 32
## - kidslt6      1 5.0111e-06 32
## - kidsge6      1 5.1174e-06 32
## - husage       1 5.2973e-06 32
## - age          1 5.7819e-06 32
## - faminc       1 6.0691e-06 32
## - nwifeinc     1 6.0865e-06 32
## - educ         1 6.4063e-06 32
## - huseduc      1 6.4724e-06 32
## - hours        1 1.1749e-05 32
## <none>         4.3679e-06 34
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
## Step:  AIC=32
## inlf ~ hours + kidslt6 + kidsge6 + age + educ + husage + huseduc +
##      huswage + faminc + motheduc + fatheduc + city + exper + nwifeinc +
##      expersq
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

```

```

## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
##           Df    Deviance AIC
## - huswage   1 4.3556e-06  30
## - fatheduc   1 4.4320e-06  30
## - motheduc   1 4.5276e-06  30
## - city       1 4.5736e-06  30
## - expersq    1 4.9102e-06  30
## - exper      1 4.9680e-06  30
## - kidslt6    1 5.0243e-06  30
## - kidsge6    1 5.0753e-06  30
## - husage     1 5.4020e-06  30
## - age        1 6.0448e-06  30
## - faminc     1 6.0813e-06  30
## - nwifeinc   1 6.0990e-06  30
## - educ       1 6.3096e-06  30
## - huseduc    1 6.7420e-06  30
## - hours      1 1.2025e-05  30
## <none>       4.3610e-06  32

```

[illegible]

```

##           Df    Deviance AIC
## - fatheduc  1 4.4674e-06  28
## - motheduc  1 4.5511e-06  28
## - city      1 4.5729e-06  28
## - expersq   1 4.9327e-06  28
## - exper     1 5.0226e-06  28
## - kidsge6   1 5.1074e-06  28
## - kidslt6   1 5.2752e-06  28
## - husage    1 5.4864e-06  28
## - age       1 6.0099e-06  28
## - huseduc   1 6.7534e-06  28
## - educ      1 7.1419e-06  28
## - nwifeinc  1 7.4272e-06  28
## - faminc    1 7.4387e-06  28
## - hours     1 1.2142e-05  28
## <none>      4.3556e-06  30

## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
## Step: AIC=28
## inlf ~ hours + kidslt6 + kidsge6 + age + educ + husage + huseduc +
##       faminc + motheduc + city + exper + nwifeinc + expersq
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge

```

```

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
##      Df    Deviance AIC
## - motheduc  1 4.5376e-06 26
## - city      1 4.7268e-06 26
## - expersq   1 5.1235e-06 26
## - exper     1 5.1694e-06 26
## - kidsge6   1 5.5157e-06 26
## - kidslt6   1 5.6296e-06 26
## - husage    1 5.8896e-06 26
## - age       1 6.5969e-06 26
## - huseduc   1 6.7504e-06 26
## - educ      1 7.3681e-06 26
## - nwifeinc  1 7.4602e-06 26
## - faminc    1 7.4822e-06 26
## - hours     1 1.5075e-05 26
## <none>      4.4674e-06 28
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
## Step: AIC=26
## inlf ~ hours + kidslt6 + kidsge6 + age + educ + husage + huseduc +
##      faminc + city + exper + nwifeinc + expersq
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge

```

```

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
##           Df    Deviance AIC
## - city      1 4.8613e-06  24
## - exper     1 5.0976e-06  24
## - expersq    1 5.1042e-06  24
## - kidsge6   1 5.5462e-06  24
## - kidslt6   1 5.7255e-06  24
## - husage    1 6.1145e-06  24
## - age       1 6.8696e-06  24
## - nwifeinc  1 7.4061e-06  24
## - faminc    1 7.4258e-06  24
## - huseduc   1 7.8554e-06  24
## - educ      1 7.8716e-06  24
## - hours     1 1.4925e-05  24
## <none>      4.5376e-06  26
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
## Step:  AIC=24
## inlf ~ hours + kidslt6 + kidsge6 + age + educ + husage + huseduc +
##       faminc + exper + nwifeinc + expersq
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

```

```

## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
##           Df    Deviance AIC
## - expersq   1 5.2235e-06  22
## - exper     1 5.3476e-06  22
## - kidslt6   1 6.0552e-06  22
## - kidsge6   1 6.3452e-06  22
## - husage    1 6.3537e-06  22
## - age       1 6.8912e-06  22
## - faminc    1 8.1254e-06  22
## - nwifeinc  1 8.1334e-06  22
## - educ      1 8.2002e-06  22
## - huseduc   1 8.3849e-06  22
## - hours     1 1.5163e-05  22
## <none>      4.8613e-06  24
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
## Step:  AIC=22
## inlf ~ hours + kidslt6 + kidsge6 + age + educ + husage + huseduc +
##       faminc + exper + nwifeinc
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge

```



```

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
##      Df    Deviance AIC
## - exper    1 5.3424e-06 20
## - kidslt6    1 6.4039e-06 20
## - kidsge6    1 6.4256e-06 20
## - husage    1 6.9718e-06 20
## - age       1 7.4512e-06 20
## - faminc    1 8.0933e-06 20
## - nwifeinc   1 8.1006e-06 20
## - huseduc    1 8.9103e-06 20
## - educ       1 9.5325e-06 20
## - hours      1 1.5793e-05 20
## <none>       5.2235e-06 22
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
## Step:  AIC=20
## inlf ~ hours + kidslt6 + kidsge6 + age + educ + husage + huseduc +
##      faminc + nwifeinc
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge

```



```

## inlf ~ hours + age + educ + huseduc + faminc + nwifeinc
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
##          Df    Deviance AIC
## - age      1 8.4436e-06  12
## - faminc    1 1.1544e-05  12
## - nwifeinc  1 1.1546e-05  12
## - educ      1 1.1826e-05  12
## - huseduc   1 1.2701e-05  12
## - hours     1 2.9251e-05  12
## <none>      8.2842e-06  14
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
## Step:  AIC=12
## inlf ~ hours + educ + huseduc + faminc + nwifeinc
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
##          Df    Deviance AIC
## - educ      1 9.2426e-06  10
## - huseduc    1 1.1116e-05  10

```

```

## - faminc      1 1.1466e-05  10
## - nwifeinc    1 1.1470e-05  10
## - hours       1 3.1571e-05  10
## <none>        8.4436e-06  12

## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
## Step: AIC=10
## inlf ~ hours + huseduc + faminc + nwifeinc
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
##           Df    Deviance AIC
## - huseduc   1 8.3090e-06   8
## - faminc    1 1.2229e-05   8
## - nwifeinc  1 1.2237e-05   8
## - hours     1 3.5583e-05   8
## <none>      9.2430e-06  10

## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
## Step: AIC=8
## inlf ~ hours + faminc + nwifeinc
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: algorithm did not converge
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
##           Df    Deviance AIC
## - faminc    1 1.0503e-05   6
## - nwifeinc  1 1.0504e-05   6
## - hours     1 3.4966e-05   6
## <none>      8.3090e-06   8

```

```

## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
## Step: AIC=6
## inlf ~ hours + nwifeinc
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
##           Df Deviance    AIC
## - nwifeinc  1      0.0    4.0
## <none>      0      0.0    6.0
## - hours    1  1019.3 1023.3
## Warning: glm.fit: algorithm did not converge

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
##
## Step: AIC=4
## inlf ~ hours
##
##           Df Deviance    AIC
## <none>      0      0.0    4.0
## - hours    1  1029.8 1031.8
summary(newmodel)

##
## Call:
## glm(formula = inlf ~ hours, family = binomial(link = logit),
##      data = data)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -1.478e-04 -1.478e-04  2.000e-08  2.000e-08  2.379e-03
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      1902      29604  0.064   0.949
## hours           2259      35082  0.064   0.949
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1.0297e+03  on 752  degrees of freedom
## Residual deviance: 1.2760e-05  on 751  degrees of freedom
## AIC: 4
##
## Number of Fisher Scoring iterations: 25

```